How Many Visual Levers Drive Urban Perception? Interventional Attribution via Localised Counterfactual Edits

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Abstract

Street-view perception models predict subjective attributes such as safety at scale, but remain correlational: they do not identify which localized visual changes would plausibly shift human judgement for a specific scene. We propose a lever-based interventional attribution framework that recasts scene-level explainability as a bounded search over structured counterfactual edits. Each lever specifies a semantic concept, spatial support, intervention direction, and constrained edit template. Candidate edits are generated through prompt-conditioned image editing and retained only if they satisfy validity checks for same-place preservation, locality, realism, and plausibility. In a pilot across 50 scenes from five cities, the framework reveals preliminary proxy-based directional patterns and a practical failure taxonomy under prompt-only editing, with Mobility Infrastructure and Physical Maintenance showing the largest auxiliary safety shifts. Human pairwise judgements remain the ground-truth endpoint for future validation.

1. Introduction

Street-view perception models predict subjective attributes such as safety, wealth, or liveliness at scale [4, 17, 21], yet remain limited as explanatory tools: they do not show which localized visual changes would plausibly shift human judgement for a specific scene. Existing explainability methods (saliency, SHAP, concept probes) remain correlational, identifying features associated with a score without validating that manipulating them changes human judgement [1, 10]. Urban-design research links cues such as greenery, maintenance, and visibility to perceived safety [9, 12, 19], but does not by itself establish which local visual changes validly shift perception in a given scene.

Following Jacobs and defensible-space theory [8, 18], we study a restricted version of this problem: whether a scene admits multiple distinct *single-lever* interventions that, evaluated independently against the same unedited original, plausibly shift perception.

We propose an *interventional attribution* protocol that makes this question testable. Rather than treating arbitrary image regions as the explanatory unit, we define structured lever interventions drawn from an editor-feasible vocabulary, each specifying a semantic concept, spatial support, intervention direction, and constrained edit template. Candidate edits are generated through prompt-conditioned diffusion editing and retained only if a vision–language critic judges them to preserve the same place, remain localised, and appear realistic and plausible. Because generative editors may introduce correlated non-target changes, each edit is treated as a hypothesis to be audited, not as faithful evidence by default.

This paper does not claim validated causal effects on human perception; it introduces an auditable framework and pilot benchmark using validity-gated generation and an auxiliary perception proxy to prioritise interventions for future human evaluation.

Our empirical questions are: *(i)* which lever families show directional improvement on an auxiliary perception proxy after validity auditing, and *(ii)* across scenes, how many distinct single-lever interventions remain promising after screening and thresholding?

Our contributions:

1. A **lever-based interventional attribution** framework for street-view perception, with the structured lever intervention $l = (c, s, d, \tau)$ as the explanatory atom.
2. A **scene-specific candidate construction and evaluation pipeline** that instantiates levers from the intervention vocabulary, realises them through constrained prompt-only edits, and audits them for same-place preservation, locality, realism, and plausibility.
3. A **reproducible pilot benchmark** across 50 scenes from five cities, characterising pipeline behaviour under prompt-only editing and reporting preliminary auxiliary patterns across lever families.

2. Related Work

Urban perception and design cues. Crowdsourced pairwise judgement datasets have enabled vision models for perceived safety, wealth, and liveliness at scale [4, 17, 20, 21], but these models are not designed to answer mechanistic scene-level questions. Urban-design theory links perceived safety to surveillance, active edges, upkeep, and legibility [8, 13, 18], and empirical street-view studies confirm roles for greenery, street-facing windows, maintenance, and lighting [3, 9, 12, 19]. However, none provides scene-level tests of localised, validity-audited single-lever interventions.

Interpretability and counterfactual explanations. Saliency and SHAP-style methods are known to be visually plausible yet unfaithful [1]; concept-based methods such as TCAV [10] remain correlational unless paired with explicit interventions. Counterfactual visual explanations [5] and causal evaluation benchmarks [14] provide stronger tools but typically target model logits rather than human perceptual preference.

Urban counterfactuals and diffusion-based editing. Law et al. [11] demonstrate plausible counterfactuals for urban image regressors, establishing that generative edits can serve as explanations in urban analytics. UrbanPhysicalDisorder-4K [16] brings counterfactual reasoning into urban safety via annotated disorder features, but operates in feature space rather than through localised image edits with explicit validity auditing. Zhao et al. [22] propose perception-guided street-view generation, but target scene optimisation rather than scene-level interventional attribution. Diffusion-based editors such as SDEdit [15], InstructPix2Pix [2], and Prompt-to-Prompt [6] make localised edits increasingly feasible, but can introduce non-target drift that demands explicit validity auditing.

Positioning. The diffusion-editing literature above optimises for editing fidelity—how faithfully a model executes an instruction—whereas our objective is *interventional attribution*: identifying which single-lever urban changes plausibly shift perception for a specific scene, with each result retained only after explicit validity auditing. The cost of prompt-only generation is imperfect locality: without a segmentation mask the editor may introduce minor non-target drift, which our critic must catch.

3. Method

3.1. Problem Setup

Let $x \in \mathcal{X}$ denote a street-view image and $a \in \mathcal{A}$ a perceptual attribute (e.g. safety). We take the explanatory atom to be a *lever intervention*

$$l = (c, s, d, \tau), \tag{1}$$

where c is a lever concept (e.g. greenery, graffiti), s is the grounded scene support, d the intervention direction (add, remove, repair), and τ a constrained edit template.

For image x, let $L(x)$ denote the scene-specific candidate set instantiated from an editor-feasible intervention vocabulary. Our method identifies the subset that can be realised as valid edits:

$$V(x) = \{l_i \in L(x) \mid \exists\, \tilde{x}_i \text{ passing the validity audit}\}. \tag{2}$$

Each lever is tested independently against the unedited original; edits are never composed sequentially.

3.2. Phase 1: Lever Candidate Construction

We define an *editor-feasible intervention vocabulary*: a bounded set of lever concepts chosen to satisfy three criteria: **(i)** *empirical grounding* in the urban perception literature [8, 9, 12, 18, 19], **(ii)** *semantic distinctness* so concepts remain interpretable, and **(iii)** *prompt-only editability* without requiring segmentation masks or major structural changes. The full concept list is given in Appendix A.3.

For each image, a VLM planner grounds applicable vocabulary concepts to visible scene regions, producing scene-specific candidates such as *remove graffiti from this wall* or *add modest greenery near this entrance*. These grounded candidates form $L(x)$.

3.3. Phase 2: Bounded Stochastic Intervention

For each candidate l_i, a prompt-only image generator G produces

$$x'_{i,j} = G(x;\, c_i, s_i, d_i, \tau_i, \varphi_j), \tag{3}$$

where φ_j indexes stochastic draws under a bounded retry budget T, with G being a prompt-only image editing model.

Each candidate is subjected to an automated validity audit performed by a vision–language model, evaluating: **(1)** same-place preservation, **(2)** locality to the intended support, **(3)** realism, and **(4)** plausibility of the intended lever. If at least one draw passes within T attempts, the first accepted edit $\tilde{x}_i$ is retained and l_i enters $V(x)$. Only edits passing all four criteria enter the retained set, so auxiliary shifts are measured only on edits with confirmed semantic and spatial integrity. We report valid-edit coverage $\mathrm{Coverage}(x) = |V(x)|/|L(x)|$ as a property of edit realisability under bounded prompt control rather than an urban perception signal.

3.4. Phase 3: Auxiliary Scoring

For each accepted edit $\tilde{x}_i$, we compute

$$\Delta_i = f_a(\tilde{x}_i) - f_a(x), \tag{4}$$

where f_a is a ViT-B/16 perception model pretrained on the MIT Place Pulse 2.0 pairwise safety dataset [7], which outputs a continuous score on a 0–10 scale (higher = safer).

This is strictly an auxiliary proxy, not the main estimand; f_a has not been validated on edited images, so Δ_i should be treated as a ranking signal rather than a calibrated effect size. The proxy-shortlisted set is

$$E_{\text{aux}}(x, a) = \{l_i \in V(x) \mid \Delta_i > \theta_{\text{aux}}\}, \tag{5}$$

where θ_{aux} is an exploratory threshold. Human-grounded pairwise evaluation remains future work.[1]

4. Preliminary Results

Dataset and setup. We use SPECS (Street Perception Evaluation Considering Socioeconomics) [20], a demographically balanced survey in which 1,000 participants from five countries rated street-view scenes on ten perceptual indicators. We sample N=50 scenes from five SPECS cities (Amsterdam, Abuja, San Francisco, Santiago, Singapore; 10 per city), stratified by baseline safety and visual complexity to cover a broad range of starting conditions. Each scene allows up to K=5 lever candidates with a stochastic budget of T=5 generation attempts per candidate. We instantiate the framework with Qwen-Image-Edit as the prompt-only editor and GPT-5.4 as the LLM-as-judge critic, yielding a reproducible lower-bound baseline. Validity auditing acts as an eligibility gate throughout: all reported effect summaries are computed only on edits retained in $V(x)$.

Auxiliary directional shifts. Among the 177 valid edits, the auxiliary scorer produces a mean shift of $+0.366$ (95% CI $[+0.199, +0.537]$, median $+0.184$, range $[-3.624, +3.842]$). Using threshold θ_{aux}=0.1, 95 of 177 valid edits fall in $E_{\text{aux}}(x, a)$, corresponding to a mean of 1.90 proxy-shortlisted levers per scene (95% CI $[1.476, 2.324]$); 40 of the 50 scenes retain at least one θ_{aux}=0.1 proxy-shortlisted lever and 24 retain multiple. The largest retained auxiliary shifts are observed for lane-marking repainting, crosswalk repainting, and localized greenery addition.

Mobility Infrastructure yields the largest family-level mean auxiliary shift ($+0.579$), led by crosswalk and lane-marking repainting. Physical Maintenance is the broadest consistently positive family ($+0.344$), driven by surface cleaning, litter removal, facade repair, and graffiti removal. Environmental Amenity is more heterogeneous: localized greenery addition is strongly positive ($+0.650$), whereas lighting repair and tree canopy management are weak or negative. Visual Legibility is small-sample and mixed in sign; the negative mean for signage decluttering (-0.631) may reflect the proxy associating commercial signage density with activity in Place Pulse training data. Per-lever proposal and valid-rate diagnostics are in Appendix Tables A1 and A2.

[1]Our method is constrained interventional search, not pixel-level decomposition or causal identification. "Multiple levers" means distinct interventions that each pass validity checks independently; it does not imply additive effects. Locality is audited, not pixel-guaranteed.

Table 1. Lever-type results grouped by intervention family. Mean Δ_{aux} and Δ_{aux} [95% CI] are computed over valid edits only; confidence intervals are bootstrap percentile intervals over the retained set. "–" denotes concepts not proposed in this scene pool.

Family / Lever concept	Valid	Mean Δ_{aux}	Δ_{aux} [95% CI]
Physical Maintenance			
Graffiti removal	4	$+0.396$	$[+0.121, +0.670]$
Litter removal	25	$+0.296$	$[-0.128, +0.743]$
Facade repair	4	$+0.519$	$[+0.105, +0.949]$
Surface cleaning	38	$+0.352$	$[+0.026, +0.684]$
Family total	71	$+0.344$	$[+0.107, +0.584]$
Environmental Amenity			
Localized greenery add.	32	$+0.650$	$[+0.241, +1.070]$
Lighting repair	8	-0.232	$[-0.598, +0.173]$
Tree canopy management	14	-0.245	$[-0.952, +0.446]$
Family total	54	$+0.287$	$[-0.041, +0.610]$
Visual Legibility			
Signage decluttering	5	-0.631	$[-1.294, -0.151]$
Storefront transparency	2	$+0.958$	$[+0.184, +1.732]$
Family total	7	-0.177	$[-0.927, +0.599]$
Mobility Infrastructure			
Crosswalk repainting	15	$+0.767$	$[+0.255, +1.261]$
Lane marking repainting	30	$+0.485$	$[+0.045, +0.934]$
Family total	45	$+0.579$	$[+0.244, +0.923]$
Overall	177	$+0.366$	$[+0.199, +0.537]$

Table 2. City-level summary over the full N=50 run. Mean Δ_{aux} and Δ_{aux} [95% CI] are computed over valid edits only. Coverage diagnostics are reported in Appendix Table A2.

City	Valid	Mean Δ_{aux}	Δ_{aux} [95% CI]
Amsterdam	40	$+0.400$	$[+0.073, +0.764]$
Abuja	32	-0.136	$[-0.441, +0.141]$
San Francisco	36	$+0.704$	$[+0.285, +1.105]$
Santiago	38	$+0.742$	$[+0.398, +1.089]$
Singapore	31	-0.012	$[-0.383, +0.352]$
Overall	177	$+0.366$	$[+0.199, +0.537]$

Thresholded proxy-shortlisted lever counts. Appendix Figure A2 counts levers that are both valid and directionally notable at increasingly strict auxiliary thresholds. At θ_{aux}=0.1, Physical Maintenance averages 0.76 proxy-shortlisted levers per scene, Mobility Infrastructure 0.56, and Environmental Amenity 0.54; Visual Legibility averages only 0.04. By city, San Francisco and Santiago dominate the positive tail with 2.5 and 2.6 proxy-shortlisted levers per scene respectively, while Abuja averages 1.1. As the cutoff rises to 1.0, the family ranking compresses but Environmental Amenity and Mobility Infrastructure retain the largest positive tails.

Santiago and San Francisco show the largest positive mean shifts; Abuja and Singapore are near zero or slightly negative. Valid-rate diagnostics by city are in Appendix Table A2.

Baseline score and gate-qualified editability. Appendix

Figure A3 plots each scene's baseline safety score $f_a(x)$ against its valid lever count $|V(x)|$. At N=50, no meaningful monotonic relationship is observed (Spearman ρ=0.10, p=0.511): scenes with lower baseline safety do not admit more valid levers than higher-baseline scenes. We revisit this null pattern in §5.

Validity gate. Of 250 proposed candidates, 177 pass the validity audit (rate 0.708); all 50 scenes retain at least one valid lever. Detailed coverage and failure analysis are in Appendix A.2.

Qualitative examples. Three positive auxiliary-shift examples – lane-marking repainting in Amsterdam, facade repair in Singapore, and surface cleaning in Santiago – are shown in Appendix Figure A1. All three are spatially constrained, visually legible, and plausible as single-lever urban changes.

5. Discussion and Future Work

This paper establishes a pilot baseline with two outputs: a feasibility profile for the generation-and-audit pipeline, and preliminary proxy-based directional patterns to prioritise for human evaluation. On feasibility, roadway-marking and maintenance interventions realise reliably across cities while lighting repair remains the clearest bottleneck (valid rate 0.30), and the null baseline-score relationship (Spearman ρ=0.10) confirms that multi-lever richness is driven by scene geometry rather than starting safety level. On directionality, Mobility Infrastructure produces the largest retained auxiliary shifts; whether this reflects human perceptual sensitivity, scorer priors, or edit realizability remains open.

The heterogeneity within Environmental Amenity is similarly informative: localised greenery addition is strongly positive (+0.650) with near-perfect validity, while lighting repair and canopy management are negative on average despite being theory-grounded cues. This pattern suggests that family-level aggregation may obscure important lever-specific differences in editability and scorer response, motivating finer-grained prioritization in future iterations of the planner.

On the auxiliary proxy. The scorer f_a has not been validated on edited images, so Δ_i should not be read as a calibrated effect size. However, for monotone interventions (road markings appear, graffiti is removed, greenery is added), the directional signal is grounded in large-scale crowdsourced pairwise preference over this class of scene variation, making the sign and relative magnitude of Δ_{aux} a useful prioritisation signal for selecting which levers and scenes to send to human evaluation.

XAI contribution. Beyond urban perception, the methodological contribution is a stronger standard for visual explanation: the proposed change must be semantically coherent and generatively feasible before it enters the attribution result, providing a reusable scaffold for interventional XAI in other perceptual domains.

Future directions and human endpoint. The present pipeline is intentionally minimal and workshop-stage: prompt-only generation, a prompted VLM critic, and an auxiliary proxy scorer are used here to establish a reproducible lower-bound baseline rather than a fully validated explanation system. Four directions follow naturally from the pipeline structure. *(i) Planner tuning:* learning to predict lever viability from scene features, using the validity and Δ_{aux} outcomes reported here as supervision, so that the K-candidate budget is spent on levers most likely to be both realisable and promising under the auxiliary proxy. *(ii) Generator alignment:* optimising for validity-gate criteria via reward-from-feedback or mask-constrained generation; the lighting-repair failure mode (valid rate 0.30) is the clearest target, with per-family valid rates as the benchmark metric. *(iii) Critic calibration:* aligning the VLM critic with human raters and ultimately learning a critic that predicts 2AFC preference directly, reducing the gap between the auxiliary proxy and the human endpoint. *(iv) End-to-end optimisation:* jointly training planner, generator, and critic once each component is independently validated. The ground-truth estimand throughout is pairwise human safety ratings on passing edits, collected via a randomised 2AFC study. Until then, validity should be read as an eligibility control on image generation and auxiliary shifts as a prioritisation signal for which lever families and cities to send to human evaluation first.

6. Conclusion

The pilot across 50 scenes demonstrates that structured, validity-audited single-lever edits can be generated at scale and scored by an auxiliary proxy, producing preliminary directional patterns – Mobility Infrastructure and Physical Maintenance yield the most consistent positive shifts – that prioritise where human evaluation should focus. More broadly, the framework shifts the explanatory unit from post-hoc pixel attribution to semantically grounded interventions whose validity is auditable before any perceptual claim is made. Realising that potential requires planner triage, stronger editing control, critic calibration, and ultimately pairwise human judgement as the ground-truth endpoint.

References

[1] Julius Adebayo, Justin Gilmer, Michael Christoph Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, pages 9505–9515, 2018. 1, 2

[2] Tim Brooks, Aleksander Holynski, and Alexei A. Efros. Instructpix2pix: Learning to follow image editing instructions.

In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. 2

[3] Marco De Nadai, Radu-Laurentiu Vieriu, Gabriele Zen, Suzana Dragicevic, Nikhil Naik, Michele Caraviello, Cesar A. Hidalgo, Nicu Sebe, and Bruno Lepri. Are safer looking neighborhoods more lively? a multimodal investigation into urban life. In *Proceedings of the 24th ACM International Conference on Multimedia*, 2016. 2

[4] Abhimanyu Dubey, Nikhil Naik, Devi Parikh, Ramesh Raskar, and Cesar A. Hidalgo. Deep learning the city: Quantifying urban perception at a global scale. In *Computer Vision – ECCV 2016*, pages 196–212. Springer, 2016. 1, 2

[5] Yash Goyal, Ziyan Wu, Jan Ernst, Dhruv Batra, Devi Parikh, and Stefan Lee. Counterfactual visual explanations. In *Proceedings of the 36th International Conference on Machine Learning*, pages 2376–2384, 2019. 2

[6] Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross-attention control. In *International Conference on Learning Representations*, 2023. 2

[7] Yujun Hou, Matias Quintana, Maxim Khomiakov, Winston Yap, Jiani Ouyang, Koichi Ito, Zeyu Wang, Tianhong Zhao, and Filip Biljecki. Global streetscapes – a comprehensive dataset of 10 million street-level images across 688 cities for urban science and analytics. *ISPRS Journal of Photogrammetry and Remote Sensing*, 215:216–238, 2024. 2

[8] Jane Jacobs. *The Death and Life of Great American Cities*. Random House, 1961. 1, 2, 10

[9] Bin Jiang, Cecilia Nga Sze Mak, Hua Zhong, Linda Larsen, and Christopher John Webster. From broken windows to perceived routine activities: Examining impacts of environmental interventions on perceived safety of urban alleys. *Frontiers in Psychology*, 9:2450, 2018. 1, 2, 10

[10] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (tcav). In *Proceedings of the 35th International Conference on Machine Learning*, pages 2668–2677, 2018. 1, 2

[11] Stephen Law, Rikuo Hasegawa, Brooks Paige, Chris Russell, and Andrew Elliott. Explaining holistic image regressors and classifiers in urban analytics with plausible counterfactuals. *International Journal of Geographical Information Science*, 37:2575–2596, 2023. 2

[12] Xiaojiang Li, Chuanrong Zhang, and Weidong Li. Does the visibility of greenery increase perceived safety in urban areas? evidence from the place pulse 1.0 dataset. *ISPRS International Journal of Geo-Information*, 4(3):1166–1183, 2015. 1, 2, 10

[13] Linda J. Loewen, G. Daniel Steel, and Peter Suedfeld. Perceived safety from crime in the urban environment. *Journal of Environmental Psychology*, 13(4):323–331, 1993. 2

[14] Thomas Melistas, Nikos Spyrou, Nefeli Gkouti, Pedro Sanchez, Athanasios Vlontzos, Yannis Panagakis, Giorgos Papanastasiou, and Sotirios A. Tsaftaris. Benchmarking counterfactual image generation. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024. 2

[15] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. Sdedit: Guided image synthesis and editing with stochastic differential equations. In *International Conference on Learning Representations*, 2022. 2

[16] Felipe A. Moreno, Andres De La Puente, and Jorge Poco. Urbanphysicaldisorder-4k: Understanding urban perception via counterfactuals and street view signs of physical disorder. In *IEEE International Conference on Big Data*, pages 5194–5200, 2025. 2

[17] Nikhil Naik, Jade Philipoom, Ramesh Raskar, and Cesar A. Hidalgo. Streetscore – predicting the perceived safety of one million streetscapes. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, pages 779–785, 2014. 1, 2

[18] Oscar Newman. *Defensible Space: Crime Prevention through Urban Design*. Macmillan, New York, 1972. 1, 2, 10

[19] Boris A. Portnov, Rasha Saad, Tal Trop, Doron Kliger, and Anna Svechkina. Linking nighttime outdoor lighting attributes to pedestrians' feeling of safety: An interactive survey approach. *PLOS ONE*, 15(11):e0242172, 2020. 1, 2, 10

[20] Matias Quintana, Youlong Gu, Xiucheng Liang, Yujun Hou, Koichi Ito, Yihan Zhu, Mahmoud Abdelrahman, and Filip Biljecki. Global urban visual perception varies across demographics and personalities. *Nature Cities*, 2(11):1092–1106, 2025. 2, 3, 10

[21] Philip Salesses, Katja Schechtner, and Cesar A. Hidalgo. The collaborative image of the city: Mapping the inequality of urban perception. *PLOS ONE*, 8(7):e68400, 2013. 1, 2

[22] Chenbo Zhao, Yoshiki Ogawa, Shenglong Chen, Takuya Oki, and Yoshihide Sekimoto. Street space quality improvement: Fusion of subjective perception in street view image generation. *Information Fusion*, 125:103467, 2026. 2

A. Appendix

A.1. Supplementary Figures

Figure A1. Three qualitative examples from the retained set. From top: lane-marking repainting (Amsterdam), facade repair (Singapore), surface cleaning (Santiago).

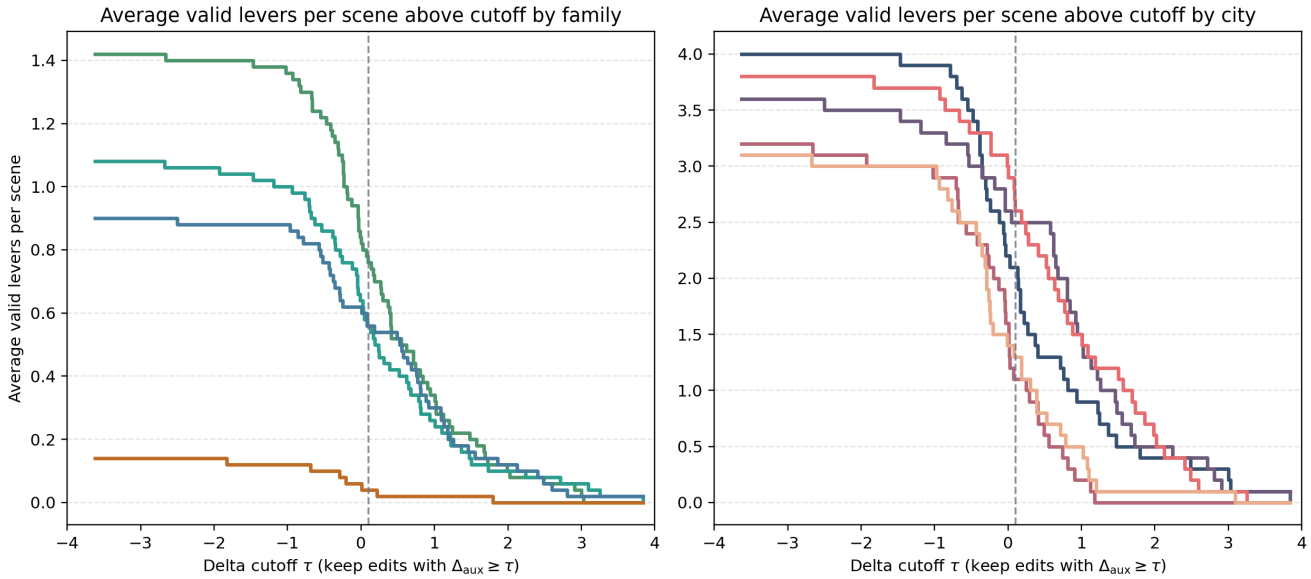

Figure A2. Average number of retained levers per scene satisfying $\Delta_{\text{aux}} \geq \tau$, split by lever family (left) and city (right). The annotation reports the overall mean Δ_{aux} with 95 % CI and the mean number of θ_{aux}=0.1 proxy-shortlisted levers per scene with 95 % CI.

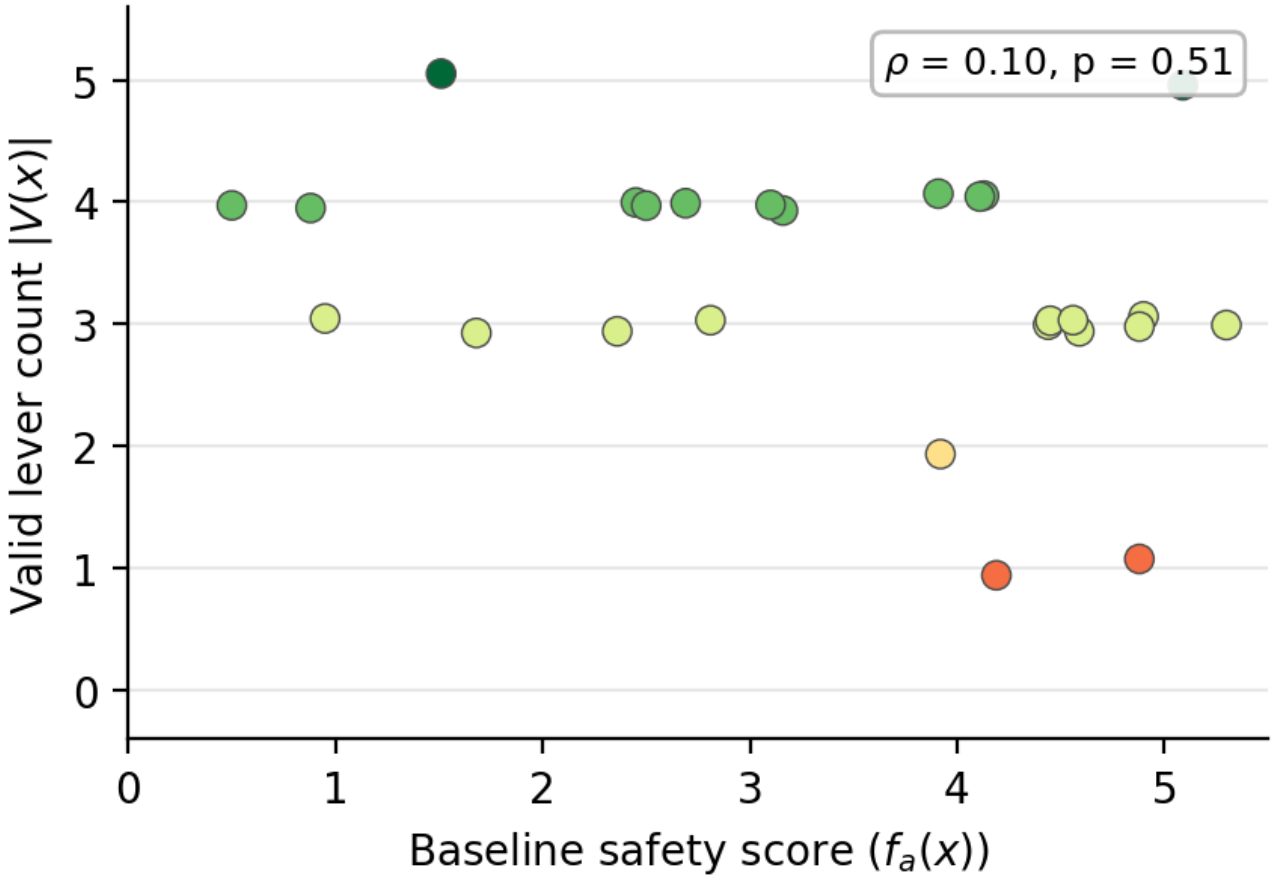

Figure A3. Baseline safety score vs. valid lever count per scene (N=50). No meaningful relationship is observed (Spearman ρ=0.10, p=0.511).

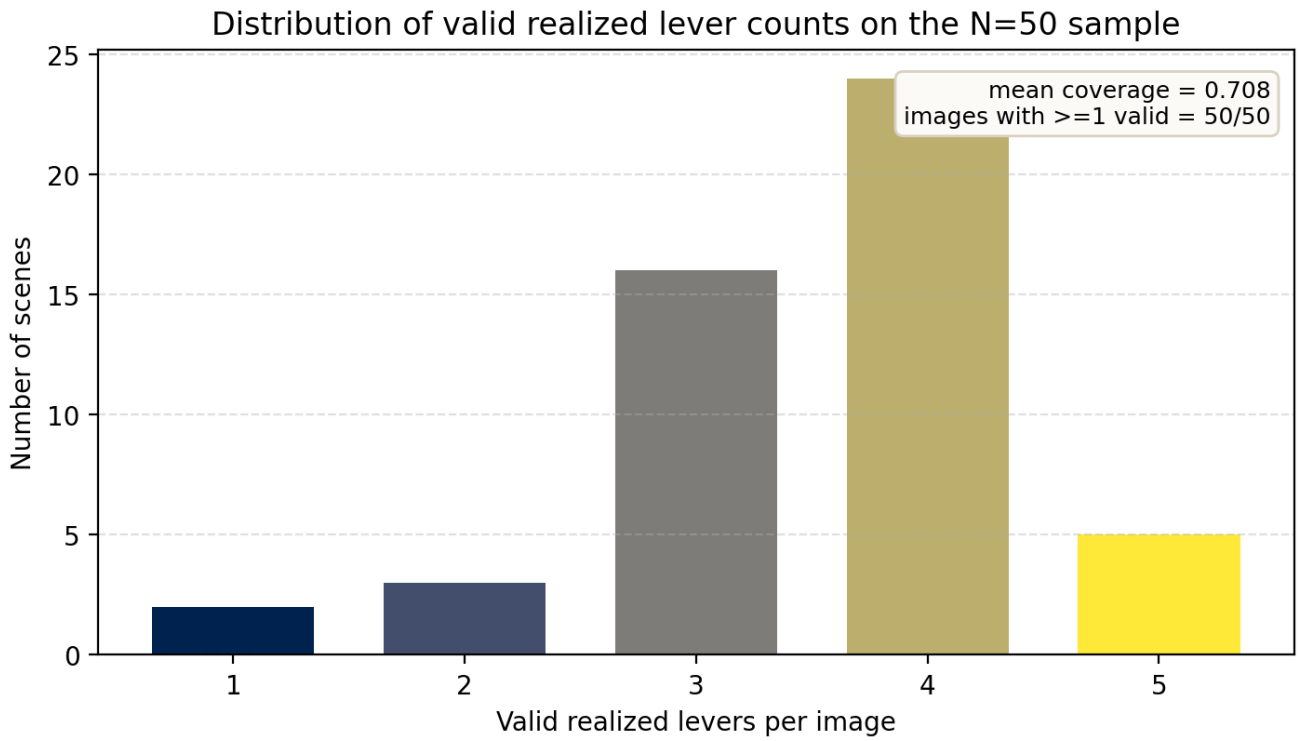

Figure A4. Distribution of valid realised lever counts per scene (N=50). Most scenes admit three or four independently valid single-lever interventions.

A.2. Validity and Coverage Details

Across 50 scenes the planner produces the full 250 candidate rows (5.0 per image on average), of which 177 pass the validity audit, yielding mean coverage $|V(x)|/|L(x)| = 0.708$. Figure A4 shows the distribution of valid counts per scene: 2 scenes have one valid lever, 3 have two, 16 have three, 24 have four, and 5 have all five.

The dominant bottleneck is not candidate proposal but valid realisation. Among the 73 audited-but-rejected edits, plausibility dominates completely; 45 also show no discernible target change, 25 exhibit non-local drift, 5 are unrealistic, and none fail same-place preservation. Generator failure before audit is zero in the cleaned N=50 run, so the remaining variance is concentrated in valid realisation rather than missing candidate production.

Table A1. Family-level summary over the full N=50 run. Valid-rate intervals are 95 % Wilson confidence intervals; mean Δ_{aux} and Δ_{aux} [95 % CI] are computed over valid edits only.

Family	Valid / Prop.	Rate [95 % CI]	Mean Δ_{aux}	Δ_{aux} [95 % CI]
Physical Maintenance	71 / 92	0.77 [0.68, 0.85]	+0.344	[+0.107, +0.584]
Environmental Amenity	54 / 82	0.66 [0.55, 0.75]	+0.287	[−0.041, +0.610]
Visual Legibility	7 / 10	0.70 [0.40, 0.89]	−0.177	[−0.927, +0.599]
Mobility Infrastructure	45 / 66	0.68 [0.56, 0.78]	+0.579	[+0.244, +0.923]
Overall	177 / 250	0.71 [0.65, 0.76]	+0.366	[+0.199, +0.537]

Table A2. City-level coverage and directional summary over the full $N{=}50$ run. Each city contributes 10 scenes and therefore 50 proposed candidates. Valid-rate intervals are 95 % Wilson confidence intervals; mean Δ_{aux} and Δ_{aux} [95 % CI] are computed over valid edits only.

City	Valid / Prop.	Rate [95 % CI]	Mean Δ_{aux}	Δ_{aux} [95 % CI]
Amsterdam	40 / 50	0.80 [0.67, 0.89]	$+0.400$	$[+0.073, +0.764]$
Abuja	32 / 50	0.64 [0.50, 0.76]	-0.136	$[-0.441, +0.141]$
San Francisco	36 / 50	0.72 [0.58, 0.83]	$+0.704$	$[+0.285, +1.105]$
Santiago	38 / 50	0.76 [0.63, 0.86]	$+0.742$	$[+0.398, +1.089]$
Singapore	31 / 50	0.62 [0.48, 0.74]	-0.012	$[-0.383, +0.352]$
Overall	177 / 250	0.71 [0.65, 0.76]	$+0.366$	$[+0.199, +0.537]$

A.3. Intervention Vocabulary

Each concept below satisfies all three inclusion criteria defined in §3. Structural interventions (building setback, road geometry, facade cladding) and social-order cues (pedestrian density, active frontage) are excluded because they are not realisable under prompt-only editing without masking or inpainting.

Physical Maintenance (broken-windows and perceived-upkeep literature [9, 18]): graffiti removal; litter removal; facade repair; surface cleaning; shutter repair.

Environmental Amenity (biophilic safety and nighttime visibility [12, 19]): localised greenery addition; lighting repair; tree canopy management.

Visual Legibility (active-edges and surveillance theory [8]): signage decluttering; storefront transparency increase.

Mobility Infrastructure (pedestrian safety and walkability [20]): crosswalk repainting; lane marking repainting.

A.4. Prompt Contracts

The planner, editor, and critic each operate under a structured prompt contract. Condensed versions are shown below; full prompts are in the released repository.

Planner prompt (condensed)

```
You are an urban perception planner.  Given a street-view image and target percept, propose a constrained set of
candidate lever interventions.

ONTOLOGY: Choose only from four families --- Physical Maintenance, Environmental Amenity, Visual Legibility,
Mobility Infrastructure --- and prefer cross-family diversity when the scene supports it.

HARD CONSTRAINTS: (1) One lever per candidate.  (2) Grounded in a visible scene element.  (3) Local, plausible,
prompt-only friendly.  (4) No global relighting, weather, or camera changes.  (5) Prefer the smallest plausible
intervention.  (6) Exclude theoretically relevant levers whose target element is not clearly visible/editable in
the image.

DIVERSITY: Return distinct candidates using different lever concepts and avoid magnitude variants of the same
intervention.

Return JSON with field ``candidates'', each containing:  lever_concept, lever_family, scene_support, target_object,
intervention_direction, edit_template, edit_plan.
```

Edit prompt (condensed)

```
Use the PROVIDED image as base.  Preserve exact viewpoint, geometry, and layout.

ALLOWED: Only modify the target object as required by the plan.  If repainting/retexturing, keep shape and
placement identical.

FORBIDDEN: No global restyling, relighting, or recoloring.  No adding/removing other objects.  No readable text.
No background, sky, road, or context changes.

Lever concept:  {lever_concept}
Lever family:  {lever_family}
Scene support:  {scene_support}
Intervention direction:  {intervention_direction}
Edit template:  {edit_template}
Target object:  {target_object}
Edit plan:  {edit_plan}
```

Critic prompt (condensed)

```
Evaluate whether the edited image is a valid single-lever counterfactual relative to the original.

CONTEXT: Edit produced by prompt-only diffusion without masking.  Minor incidental changes (tone drift, texture
resampling) are expected artefacts and should not cause failure unless they materially alter scene meaning;
plausibility and locality are still judged strictly against the requested lever and support.

CRITERIA: (A) edit_attempted:  generator made a visible change at the target (false only if output looks identical
to original there).  (B) same_place_preserved:  same underlying place.  (C) is_localised:  primary meaningful change
is in/near intended support; minor global tone shifts do not count as non-local.  (D) is_realistic:  physically
plausible and coherent.  (E) is_plausible:  recognisably the requested lever at the stated support; fail if the
edit type/support is wrong or too excessive.

CLEAR FAIL CONDITIONS: Viewpoint/geometry changed; large non-target objects added/removed; requested lever
replaced by a different change; no discernible change at target.

Return JSON: {edit_attempted, same_place_preserved, is_localised, is_realistic, is_plausible, notes}, where notes =
{failure_modes, diagnosis, repair_suggestion}.
```